UNIT 2 · ARS/NET PATTERN

MCQ Bank — 215 Questions with Answers

Machinery elements — Topics 1–2 (Q1–Q40)

Q1. Power transmitted by a shaft is given by:

(a) 2πNT/60 W (b) πNT/60 W
(c) NT/60 W (d) 2NT/π W

Q2. In a belt drive, the ratio T_1/T_2 equals:

(a) e^kθ (b) kθ
(c) e^θ/k (d) log(kθ)

Q3. The tension ratio commonly assumed in designing V-belt drives is:

(a) 2 (b) 3
(c) 5 (d) 8

Q4. For rubber flat belts on steel pulleys, the tension ratio at 180° arc of contact is about:

(a) 1.2–1.5 (b) 2.2–2.6
(c) 4–5 (d) 6–8

Q5. The included angle of a standard V-belt cross-section is:

(a) 30° (b) 34°
(c) 38° (d) 40°

Q6. Sheave groove angles are kept at 34–38°, i.e., less than the belt angle, to:

(a) Reduce belt speed (b) Ensure wedging action
(c) Increase slip (d) Ease belt removal

Q7. A V-belt sheave and a flat pulley are balanced in power capacity when arcs of contact are about:

(a) 180° and 180° (b) 130° and 230°
(c) 230° and 130° (d) 90° and 270°

Q8. Belt power is maximum when centrifugal tension equals:

(a) T_1 (b) $T_1/2$
(c) $T_1/3$ (d) $3T_1$

Q9. Creep in belts is caused by:

(a) Overload slip (b) Alternate elastic elongation/contraction of the belt over pulleys
(c) Misalignment (d) Wear

Q10. Agricultural V-belt cross-sections per ASAE standards are designated:

(a) A, B, C (b) HA, HB, HC, HD, HE
(c) SPA, SPB (d) 3V, 5V, 8V

Q11. Double (hexagonal) V-belts are used for:

(a) Higher speeds (b) Serpentine drives with reverse bends
(c) Vertical shafts (d) Wet conditions

Q12. Flat-belt pulleys are crowned to:

(a) Increase friction
(b) Keep the belt tracking at pulley centre
(c) Reduce weight
(d) Balance the pulley

Q13. An idler pulley placed on the slack side near the small pulley:

(a) Reduces arc of contact
(b) Increases arc of contact and capacity
(c) Increases slip
(d) Reduces belt life only

Q14. Variable-speed V-belt drives change speed by:

(a) Changing motor speed
(b) Adjustable-pitch sheaves altering effective pitch diameter
(c) Gearbox
(d) Slipping clutch

Q15. Chordal (polygonal) action in chain drives causes:

(a) Chain elongation
(b) Cyclic variation in chain velocity
(c) Wear of pins
(d) Sprocket eccentricity

Q16. To minimize chordal action, the smaller sprocket should generally have at least:

(a) 6 teeth
(b) 10 teeth
(c) 17 teeth
(d) 30 teeth

Q17. The chain best suited to gritty, slow-speed farm conveyor service and field repair is:

(a) Silent chain
(b) Precision roller chain
(c) Detachable-link (steel/malleable) chain
(d) Leaf chain

Q18. Chain speed (m/s) with pitch p (m), sprocket teeth T, speed N (rpm) is:

(a) pTN/60
(b) pT/60N
(c) 60pTN
(d) pN/60T

Q19. Unlike belts, chain drives are operated with:

(a) High initial tension
(b) A small amount of slack
(c) Zero backlash
(d) Crowned sprockets

Q20. Attachment links (A- and K-type) on roller chains are used in agriculture to:

(a) Join chain ends
(b) Mount slats/paddles on conveyor/elevator chains
(c) Reduce noise
(d) Increase strength

Q21. The standard pressure angle for involute gears today is:

(a) 14.5°
(b) 20°
(c) 25°
(d) 30°

Q22. Minimum number of teeth on a 20° full-depth pinion to avoid interference is about:

(a) 12
(b) 17
(c) 25
(d) 32

Q23. Module of a gear equals:

(a) T/D
(b) D/T
(c) πD/T
(d) T/πD

Q24. The gear pair connecting shafts whose axes are offset (non-intersecting, non-parallel), used in automotive/tractor drive lines, is:

(a) Spiral bevel
(b) Hypoid
(c) Herringbone
(d) Rack and pinion

Q25. A worm-and-wheel drive is preferred where:

(a) High efficiency is needed
(b) Large speed reduction in one stage (and possible self-locking) is needed
(c) Shafts are parallel
(d) Axial thrust must be zero

Q26. In the Lewis equation $F_t = \sigma \cdot b \cdot Y \cdot m$, Y is the:

(a) Wear factor
(b) Lewis form factor (tooth shape)
(c) Service factor
(d) Velocity factor

Q27. An idler gear in a simple gear train:

(a) Changes speed ratio
(b) Changes only the direction of rotation
(c) Increases torque
(d) Reduces noise

Q28. L_{10} (B_{10}) bearing life is the life that:

(a) 10% of bearings complete
(b) 90% of a group of bearings will complete or exceed
(c) Average bearing achieves
(d) The best bearing achieves

Q29. Bearing life varies with load as:

(a) $L \propto C/P$
(b) $L \propto (C/P)^3$ for ball bearings
(c) $L \propto (C/P)^2$
(d) L ∝ e^(C/P)

Q30. For combined heavy radial and axial loads in tractor wheel hubs, the bearing used is:

(a) Deep-groove ball
(b) Needle roller
(c) Tapered roller
(d) Self-aligning ball

Q31. Universal-joint cross trunnions usually run in:

(a) Bush bearings
(b) Needle roller bearings
(c) Ball bearings
(d) Tapered rollers

Q32. Speed fluctuation of a single Hooke's joint is eliminated in a PTO shaft by:

(a) Higher speed
(b) Two joints with in-phase yokes and equal joint angles
(c) Heavier shaft
(d) Lubrication

Q33. Oldham coupling accommodates:

(a) Angular misalignment
(b) Parallel (lateral) shaft offset
(c) Axial float only
(d) Torsional vibration

Q34. The overload device that gives an audible warning and re-engages automatically is the:

(a) Shear pin
(b) Jump (ratchet) clutch
(c) Rigid coupling
(d) Key

Q35. An overrunning (freewheel) clutch in a rotavator/PTO driveline:

(a) Limits torque
(b) Allows the high-inertia rotor to overrun the PTO while slowing
(c) Reverses drive
(d) Increases speed

Q36. Recommended spring index (C = D/d) for helical springs is:

(a) 1–3
(b) 6–12
(c) 15–25
(d) 30–50

Q37. Wahl factor corrects the stress in helical springs for:

(a) Temperature
(b) Curvature and direct shear effects
(c) Buckling
(d) Fatigue only

Q38. A bolt of property class 8.8 has minimum ultimate tensile strength and yield strength of:

(a) 800 and 640 MPa
(b) 880 and 800 MPa
(c) 800 and 880 MPa
(d) 640 and 800 MPa

Q39. The 540-rpm tractor PTO shaft has:

(a) 21 splines
(b) 6 splines
(c) 8 splines
(d) 10 splines

Q40. The self-aligning key seated in a semicircular shaft recess is the:

(a) Feather key
(b) Kennedy key
(c) Woodruff key
(d) Gib-head key

Answers

Q1 — (a) 2πNT/60 W

Q2 — (a) e^kθ

Q3 — (c) 5 — *Kepner §4.3 — flat rubber belt k = 0.25–0.30 gives R = 2.2–2.6 at 180°; V-belt design assumes R = 5 (k ≈ 0.51 including wedging).*

Q4 — (b) 2.2–2.6 — *Kepner §4.3 — flat rubber belt k = 0.25–0.30 gives R = 2.2–2.6 at 180°.*

Q5 — (d) 40°

Q6 — (b) Ensure wedging action

Q7 — (b) 130° and 230° — *Equal kθ values → 130° sheave ≈ 230° flat pulley (Kepner §4.9); basis of V-flat drives on threshers.*

Q8 — (c) $T_1/3$ — *dP/dv = 0 with P = (T_1 -T_c-T_2')v … yields T_c = T_1/3.*

Q9 — (b) Alternate elastic elongation/contraction of the belt over pulleys

Q10 — (b) HA, HB, HC, HD, HE

Q11 — (b) Serpentine drives with reverse bends

Q12 — (b) Keep the belt tracking at pulley centre

Q13 — (b) Increases arc of contact and capacity

Q14 — (b) Adjustable-pitch sheaves altering effective pitch diameter

Q15 — (b) Cyclic variation in chain velocity

Q16 — (c) 17 teeth — *Chordal velocity variation ∝ (1 - cos 180°/T); below 17 teeth the fluctuation and noise rise steeply.*

Q17 — (c) Detachable-link (steel/malleable) chain

Q18 — (a) pTN/60

Q19 — (b) A small amount of slack

Q20 — (b) Mount slats/paddles on conveyor/elevator chains

Q21 — (b) 20°

Q22 — (b) 17 — *Minimum teeth = 2/sin²20° ≈ 17.1 → 17–18.*

Q23 — (b) D/T

Q24 — (b) Hypoid

Q25 — (b) Large speed reduction in one stage (and possible self-locking) is needed

Q26 — (b) Lewis form factor (tooth shape)

Q27 — (b) Changes only the direction of rotation

Q28 — (b) 90% of a group of bearings will complete or exceed — *L_{10} = (C/P)^p million rev; 90% reliability definition — repeatedly asked.*

Q29 — (b) L ∝ $(C/P)^3$ for ball bearings — *L_{10} = (C/P)^p million rev, with p = 3 for ball bearings.*

Q30 — (c) Tapered roller

Q31 — (b) Needle roller bearings

Q32 — (b) Two joints with in-phase yokes and equal joint angles — *PTO drivelines are double-Cardan arrangements; equal angles + in-phase yokes cancel the cos α fluctuation.*

Q33 — (b) Parallel (lateral) shaft offset
Q34 — (b) Jump (ratchet) clutch
Q35 — (b) Allows the high-inertia rotor to overrun the PTO while slowing
Q36 — (b) 6–12
Q37 — (b) Curvature and direct shear effects
Q38 — (a) 800 and 640 MPa — *Class X.Y → UTS = 100X MPa; yield = 0.Y × UTS.*
Q39 — (b) 6 splines
Q40 — (c) Woodruff key

System characteristics, balancing and stability — Topics 3–4 (Q41–Q75)

Q41. The capacity of a multi-component machine system (e.g., combine) is determined by:

(a) The largest component
(b) The limiting (weakest) functional component
(c) The engine alone
(d) Average of all components

Q42. If a machine has 4 components in series, each with reliability 0.95, system reliability is about:

(a) 0.95
(b) 0.99
(c) 0.81
(d) 0.86

Q43. Draft of a mouldboard plough varies with forward speed as:

(a) $D = a + bV$
(b) $D = a + bV^2$
(c) $D = a/V$
(d) D = constant

Q44. Quality of work of a seed drill is judged by:

(a) Fuel use
(b) Seed placement uniformity/depth accuracy
(c) Weight
(d) Draft only

Q45. RNAM test codes relate to:

(a) Tractor engines
(b) Standard test procedures for agricultural machines in Asia-Pacific region
(c) Chemical residues
(d) Irrigation design

Q46. A rotor at rest in any angular position without turning is:

(a) Dynamically balanced
(b) Statically balanced
(c) Unbalanced
(d) Over-balanced

Q47. Static balance requires:

(a) ΣForces = 0
(b) ΣCouples = 0
(c) Both zero
(d) High speed

Q48. Dynamic balance additionally requires:

(a) Zero resultant couple
(b) Higher mass
(c) One correction plane
(d) Rigid shaft

Q49. Masses balanced in one plane but distributed along the shaft may still produce:

(a) No effect
(b) A rocking (unbalanced) couple
(c) Static unbalance
(d) Torque rise

Q50. Centrifugal force due to unbalance varies as:

(a) ω
(b) ω^2
(c) ω^3
(d) $\sqrt{\omega}$

Q51. Doubling the speed of an unbalanced thresher cylinder increases the unbalance force:

(a) 2×
(b) 4×
(c) 8×
(d) No change

Q52. Complete dynamic balancing of rotating masses in several planes needs:

(a) One counterweight
(b) Correction in two planes
(c) Flywheel
(d) Damper

Q53. A reciprocating inertia force CANNOT be fully balanced by a rotating counterweight because:

(a) It is too large
(b) Rotating mass gives constant-magnitude force while reciprocating force varies with crank angle
(c) Speeds differ
(d) Friction interferes

Q54. In mower balancing practice, the rotating counterweight is sized to balance the rotating parts plus about:

(a) One-fourth
(b) One-half of maximum reciprocating force
(c) Full reciprocating force
(d) Twice the reciprocating force

Q55. In the analysis of mower pitman forces, the pitman mass is idealized as:

(a) Uniform rod
(b) Two concentrated masses at crankpin (rotating) and knife head (reciprocating)
(c) Point at centre
(d) Neglected

Q56. The primary reciprocating inertia force has a frequency equal to:

(a) Half crank speed
(b) Crank speed
(c) Twice crank speed
(d) Four times

Q57. Two opposed reciprocating counter-masses fully cancel forces only if they are:

(a) Unequal
(b) Collinear (otherwise a vibratory couple results)
(c) At right angles
(d) On separate shafts

Q58. The double-knife (opposed knives) cutter bar permits:

(a) Lower cutting speed
(b) Half stroke per knife and higher crank rpm → higher forward speed
(c) Heavier knives
(d) No guards in all versions

Q59. In a dynamically balanced mower (Kepner Fig. 14.6), the balancing element is:

(a) Heavier flywheel
(b) Reciprocating counterweight on the inner shoe
(c) Rubber mounts
(d) Longer pitman

Q60. The energy-smoothing member for the baler plunger's cyclic load is the:

(a) Knotter
(b) Flywheel
(c) Pickup
(d) Bale tension rail

Q61. Coefficient of fluctuation of speed is:

(a) $(\omega_{max}-\omega_{min})/\omega_{mean}$
(b) $\omega_{mean}/(\omega_{max}-\omega_{min})$
(c) $\omega_{max}/\omega_{min}$
(d) $\Delta\omega^2$

Q62. Required flywheel moment of inertia for energy fluctuation ΔE is:

(a) $\Delta E \cdot C_s \cdot \omega^2$
(b) $\Delta E/(C_s \cdot \omega^2_{mean})$
(c) $C_s/\Delta E$
(d) $\frac{1}{2}I\omega^2$

Q63. Resonance occurs when:

(a) Damping is high
(b) Excitation frequency equals natural frequency
(c) Mass is large
(d) Speed is low

Q64. Natural frequency of a spring-mass system is:

(a) √(m/k) (b) √(k/m)

(c) k/m (d) m/k

Q65. Vibration isolation (transmissibility <1) requires frequency ratio greater than:

(a) 1 (b) √2

(c) 2 (d) π

Q66. A machine on a side slope overturns when the vertical through the CG:

(a) Passes inside wheelbase (b) Falls outside the wheel-contact polygon

(c) Is at axle height (d) Coincides with draft line

Q67. Lateral overturn limit angle with track width B and CG height h is given by:

(a) tan θ = 2h/B (b) tan θ = B/2h

(c) sin θ = B/h (d) θ = B·h

Q68. Lateral stability is improved by:

(a) Higher CG, narrow track (b) Lower CG, wider track

(c) More speed (d) Higher hitch

Q69. With a heavy rear-mounted implement raised, safe steering requires the front axle to retain at least about:

(a) 5% (b) 20%

(c) 40% (d) 60% of total weight

Q70. Front-end ballast requirement for a mounted implement is computed by:

(a) Energy balance (b) Moment balance about the rear axle

(c) Draft equation (d) Traction equation

Q71. Positive tongue (hitch) load recommended on trailed trailers is about:

(a) 0% (b) 10–15% of gross weight

(c) 30–40% (d) 50%

Q72. Snaking/weaving of a trailed implement at transport speed is reduced by:

(a) Negative tongue load (b) Adequate tongue load and proper hitch length

(c) Higher speed (d) Wider tyres

Q73. An unbalanced high-speed cylinder in a thresher primarily causes:

(a) Better threshing (b) Vibration, bearing failure and frame cracks

(c) Higher capacity (d) Less noise

Q74. Timeliness of operation in machinery system evaluation refers to:

(a) Machine speed (b) Completing the operation within the optimum period to avoid yield/quality loss

(c) Idle time (d) Shift hours

Q75. MTBF stands for:

(a) Maximum time before failure (b) Mean time between failures

(c) Minimum test bench force (d) Mean torque at base frequency

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Answers

Q41 — (b) The limiting (weakest) functional component

Q42 — (c) 0.81 — *$0.95^4 = 0.8145$ — series reliability multiplies; why simple, robust designs matter in farm machines.*

Q43 — (b) $D = a + bV^2$

Q44 — (b) Seed placement uniformity/depth accuracy

Q45 — (b) Standard test procedures for agricultural machines in Asia-Pacific region
Q46 — (b) Statically balanced
Q47 — (a) ΣForces = 0
Q48 — (a) Zero resultant couple
Q49 — (b) A rocking (unbalanced) couple
Q50 — (b) ω²
Q51 — (b) 4× — *F = mrω² → 2² = 4×.*
Q52 — (b) Correction in two planes
Q53 — (b) Rotating mass gives constant-magnitude force while reciprocating force varies with crank angle — *Kepner §14.11 — the rotating counterweight force is constant at all crank angles.*
Q54 — (b) One-half of maximum reciprocating force — *Kepner §14.11 — compromise counterweight = F_c + ½F_r,max ("half balancing"), residual unbalance shifted 90°.*
Q55 — (b) Two concentrated masses at crankpin (rotating) and knife head (reciprocating)
Q56 — (b) Crank speed
Q57 — (b) Collinear (otherwise a vibratory couple results)
Q58 — (b) Half stroke per knife and higher crank rpm → higher forward speed — *Kepner §14.12 — each knife stroke halves; crank speed (and forward speed) can nearly double for the same inertia stress.*
Q59 — (b) Reciprocating counterweight on the inner shoe
Q60 — (b) Flywheel
Q61 — (a) (ω_max-ω_min)/ω_mean
Q62 — (b) ΔE/(C_s·ω²_mean)
Q63 — (b) Excitation frequency equals natural frequency
Q64 — (b) √(k/m)
Q65 — (b) √2 — *Transmissibility < 1 only when ω/ω_n > √2 — basis of soft engine mounts.*
Q66 — (b) Falls outside the wheel-contact polygon
Q67 — (b) tan θ = B/2h — *At tipping, the CG vertical passes through the downhill wheel contact: tan θ = (B/2)/h.*
Q68 — (b) Lower CG, wider track
Q69 — (b) 20%
Q70 — (b) Moment balance about the rear axle
Q71 — (b) 10–15% of gross weight
Q72 — (b) Adequate tongue load and proper hitch length
Q73 — (b) Vibration, bearing failure and frame cracks
Q74 — (b) Completing the operation within the optimum period to avoid yield/quality loss
Q75 — (b) Mean time between failures

Force analysis, pull–draft–power and furrow openers — Topics 5–6 (Q76–Q110)

Q76. Draft of an implement is:

(a) Total force along the hitch line
(b) Horizontal component of pull in the direction of travel
(c) Vertical component of pull
(d) Side component of pull

Q77. A pull of 3000 N acts at 20° above horizontal. The draft is about:

(a) 1026 N
(b) 2819 N
(c) 3000 N
(d) 3192 N

Q78. Unit draft is defined as draft per unit:

(a) Width of implement
(b) Weight of implement
(c) Cross-sectional area of furrow
(d) Travel speed

Q79. The unit of unit draft is:

(a) N
(b) N/cm
(c) N/cm²
(d) N·m

Q80. 1 kgf/cm² is approximately:

(a) 9.81 kPa
(b) 98.1 kPa
(c) 981 kPa
(d) 9810 kPa

Q81. A 2-bottom MB plough (30 cm/bottom) works 15 cm deep with unit draft 0.6 kgf/cm². Total draft is:

(a) 270 kgf
(b) 405 kgf
(c) 540 kgf
(d) 810 kgf

Q82. The plough in Q6 moves at 1.5 m/s. Drawbar power is about:

(a) 5.4 hp
(b) 7.2 hp
(c) 10.8 hp
(d) 14.4 hp

Q83. With D in kgf and v in km/h, drawbar power in kW equals:

(a) D·v/75
(b) D·v/270
(c) D·v/367
(d) D·v/1000

Q84. Draft 9 kN at 1.5 m/s gives drawbar power:

(a) 6 kW
(b) 9 kW
(c) 13.5 kW
(d) 60 kW

Q85. In Goryachkin's equation P = fG + kab + εabv², the speed-dependent term is:

(a) fG
(b) kab
(c) εabv²
(d) All three

Q86. The term fG in Goryachkin's equation represents:

(a) Soil cutting resistance
(b) Friction/rolling resistance due to implement weight
(c) Soil throwing
(d) Suction force

Q87. If ploughing speed is doubled, the dynamic (soil-throwing) component of draft becomes:

(a) 2×
(b) 4×
(c) 8×
(d) Unchanged

Q88. In ASAE D497 the soil-texture factor Fi for a mouldboard plough in fine (clay) soil is:

(a) 0.45
(b) 0.70
(c) 1.0
(d) 1.5

Q89. The centre of resistance of an MB plough bottom is commonly assumed at:

(a) ¼ width from landside, ½ depth
(b) ½ width, full depth
(c) At the share point
(d) At the heel of landside

Q90. For steady running with minimum draft, the line of pull must pass through:

(a) The rear axle
(b) Centre of resistance and hitch point
(c) The plough heel
(d) The mast top

Q91. Side draft occurs when:

(a) Speed is high
(b) The line of pull is laterally offset from the centre of resistance
(c) Depth is uniform
(d) Soil is sandy

Q92. Hitching a trailed plough too HIGH on the tractor drawbar generally causes:

(a) Deeper penetration
(b) Reduced penetration (plough runs shallow)
(c) No change
(d) More side draft only

Q93. A disc plough penetrates mainly because of:

(a) Suction of the bottom
(b) Added weight/ballast
(c) Sharp landside
(d) High speed

Q94. Which of these is a PARASITIC force on a plough?

(a) Soil cutting resistance
(b) Furrow slice acceleration
(c) Landside friction
(d) Soil inversion resistance

Q95. In L–V–S force analysis of a mounted implement, V denotes:

(a) Velocity
(b) Vertical force component
(c) Volume of soil
(d) Side force

Q96. Draft of a TRAILED implement is best measured with:

(a) Torque meter on PTO
(b) Dynamometer between drawbar and implement
(c) Fuel meter
(d) Tachometer

Q97. Draft of a MOUNTED implement is measured with:

(a) Spring balance on top link only
(b) Three-point hitch (strain-gauge) dynamometer
(c) Rope and pulley
(d) Prony brake

Q98. Compared with sandy soil, the unit draft of ploughing heavy clay is:

(a) Lower
(b) Same
(c) Higher
(d) Zero

Q99. Draft of a tillage implement depends on:

(a) Soil type and moisture
(b) Depth and width of cut
(c) Speed and tool geometry
(d) All of these

Q100. Draft of seed drills and harrows is usually expressed per unit:

(a) Furrow area
(b) Working width
(c) Weight
(d) Wheel diameter

Q101. Drawbar power equals engine power multiplied by:

(a) Transmission efficiency only
(b) Tractive efficiency only
(c) Transmission × tractive efficiency
(d) Field efficiency

Q102. Engine power 60 kW, transmission efficiency 0.90, tractive efficiency 0.67. Drawbar power ≈

(a) 24 kW
(b) 30 kW
(c) 36 kW
(d) 54 kW

Q103. The furrow opener best suited for zero-till sowing into anchored rice residue is:

(a) Runner
(b) Inverted-T
(c) Shovel
(d) Full-width sweep

Q104. Double-disc furrow openers are preferred on planters because they give:

(a) Maximum soil inversion
(b) Precise, uniform seed placement and cut trash
(c) Deepest furrows
(d) Lowest cost

Q105. Hoe-type furrow openers are best suited to:

(a) Wet puddled soil
(b) Hard and dry soils
(c) Loose sand only
(d) Standing water

Q106. Runner-type openers require:

(a) Trashy uneven land
(b) Clean, well-prepared moist seedbed
(c) Stony soil
(d) Deep mud

Q107. The press wheel behind a furrow opener serves to:

(a) Increase draft
(b) Firm soil around seed for better seed–soil contact and germination
(c) Meter the seed
(d) Mark the row

Q108. Disc-type openers are preferred over shoe type when:

(a) Field has heavy trash/residue
(b) Soil is clean and loose
(c) Speed is very low
(d) Seeds are very small

Q109. Increasing the rake (approach) angle of a tine well beyond its optimum causes draft to:

(a) Decrease
(b) Increase
(c) Remain constant
(d) Become zero

Q110. Spring-loaded tines on cultivators are provided mainly to:

(a) Increase working depth
(b) Protect against stones/obstructions by breakaway action
(c) Reduce cost
(d) Improve seed metering

Answers

Q76 — (b) Horizontal component of pull in the direction of travel

Q77 — (b) 2819 N — *$D = 3000 \cdot \cos 20° = 3000 \times 0.9397 \approx 2819$ N; vertical component $3000 \cdot \sin 20° \approx 1026$ N (distractor a).*

Q78 — (c) Cross-sectional area of furrow

Q79 — (c) N/cm²

Q80 — (b) 98.1 kPa

Q81 — (c) 540 kgf — *D = unit draft × furrow area × bottoms = 0.6 × (30×15) × 2 = 540 kgf.*

Q82 — (c) 10.8 hp — *$P = D \cdot v/75 = 540 \times 1.5/75 = 10.8$ hp ($\approx$ 7.95 kW).*

Q83 — (c) D·v/367

Q84 — (c) 13.5 kW — *$P(kW) = D(kN) \times v(m/s) = 9 \times 1.5 = 13.5$ kW.*

Q85 — (c) εabv^2

Q86 — (b) Friction/rolling resistance due to implement weight

Q87 — (b) 4× — *Dynamic term $\propto v^2 \rightarrow 2^2 = 4\times$. This is why high-speed ploughing needs disproportionately more power.*

Q88 — (c) 1.0

Q89 — (a) ¼ width from landside, ½ depth

Q90 — (b) Centre of resistance and hitch point

Q91 — (b) The line of pull is laterally offset from the centre of resistance

Q92 — (b) Reduced penetration (plough runs shallow) — *Higher hitch → steeper pull line → larger upward component $P \cdot \sin\theta$ on the implement → reduced penetration.*

Q93 — (b) Added weight/ballast

Q94 — (c) Landside friction

Q95 — (b) Vertical force component

Q96 — (b) Dynamometer between drawbar and implement

Q97 — (b) Three-point hitch (strain-gauge) dynamometer

Q98 — (c) Higher
Q99 — (d) All of these
Q100 — (b) Working width
Q101 — (c) Transmission × tractive efficiency
Q102 — (c) 36 kW — *60 × 0.90 × 0.67 ≈ 36.2 kW — barely 60% of engine power reaches the drawbar on soil.*
Q103 — (b) Inverted-T
Q104 — (b) Precise, uniform seed placement and cut trash
Q105 — (b) Hard and dry soils
Q106 — (b) Clean, well-prepared moist seedbed
Q107 — (b) Firm soil around seed for better seed–soil contact and germination
Q108 — (a) Field has heavy trash/residue
Q109 — (b) Increase
Q110 — (b) Protect against stones/obstructions by breakaway action

Sowing, fertilizer and inter-cultivation design — Topics 6–7 (Q111–Q145)

Q111. The target cell fill for a precision planter plate is:

(a) 50–60% (b) 70–80%
(c) 95–100% (d) Exactly 110%

Q112. As plate (cell) linear speed increases, cell fill:

(a) Increases (b) Decreases
(c) Constant (d) First falls then rises

Q113. Barmington's data (Kepner Fig. 11.18) showed 100% cell fill for sugar-beet seed at a plate speed of about:

(a) 10 fpm (b) 50 fpm
(c) 300 fpm (d) 500 fpm

Q114. To plant at higher forward speeds with a plate meter, the designer should:

(a) Speed up the plate (b) Use more cells per plate or a larger plate
(c) Reduce cells (d) Raise the hopper

Q115. The metering-unit element that prevents multiple fill and seed breakage is the:

(a) Knock-out (b) Cutoff device
(c) Agitator (d) Gate spring

Q116. Positive removal of seed from cells is ensured by:

(a) Gravity only (b) Knock-out/ejector devices
(c) Vacuum (d) Higher speed

Q117. Conditions giving 95–100% cell fill also give:

(a) Max damage (b) Most uniform spacing and minimum seed damage
(c) Higher draft (d) Poor spacing

Q118. In fluted-roller metering, seed rate is chiefly controlled by:

(a) Roller diameter (b) Exposed flute length
(c) Seed tube length (d) Hopper level

Q119. Helical flutes (vs straight) on a fluted roller give:

(a) Higher damage (b) Steadier, more uniform seed flow
(c) Lower capacity only (d) More wear

Q120. Seed spacing delivered by a planter (D_w wheel dia, σ slip, n_c cells, i plate/wheel ratio) is:

(a) $\pi D_w(1-\sigma)/(n_c \cdot i)$ (b) $n_c \cdot i/\pi D_w$

(c) $\pi D_w \cdot n_c \cdot \sigma$ (d) D_w/n_c

Q121. Ground-wheel slip (skid) of a drill causes actual seed spacing to be:

(a) Narrower than theoretical (b) Wider than theoretical

(c) Unchanged (d) Random

Q122. Hopper walls must slope:

(a) Less than angle of repose (b) Greater than the angle of repose of the material

(c) Exactly 30° (d) Any angle

Q123. The angle between the two discs of a double-disc furrow opener is about:

(a) 2–3° (b) 7–10°

(c) 20–25° (d) 40°

Q124. Heavy down-force springs (50–150 kgf/opener) are characteristic of:

(a) Broadcasting (b) No-till/zero-till openers

(c) Cup feed drills (d) Check-row planters

Q125. A press wheel behind the opener serves to:

(a) Mark rows (b) Firm soil around seed for better seed–soil contact

(c) Drive the meter (d) Cut trash

Q126. Fertilizer metering is inherently difficult because fertilizers are:

(a) Uniform and dry (b) Hygroscopic, corrosive, variable in flowability

(c) Dense only (d) Fibrous

Q127. The feed used extensively on grain-drill fertilizing units is:

(a) Vacuum disc (b) Star-wheel feed

(c) Cell plate (d) Spinning disc

Q128. In star-wheel feed, application rate is adjusted by:

(a) Fan speed (b) Gate opening and wheel size/speed

(c) Boom height (d) Nozzle size

Q129. Auger (screw) feeder delivery rate is proportional to:

(a) Screw speed and pitch volume (b) Hopper height

(c) Gate colour (d) Tube length

Q130. The single-pass distribution pattern of a spinning-disc broadcaster is:

(a) Rectangular (b) Triangular/oval, requiring 25–35% overlap

(c) Ring (d) Square

Q131. In spinner spreading, fine particles:

(a) Travel farthest (b) Fall short — causing pattern segregation

(c) Do not move (d) Bounce back

Q132. Particle velocity leaving a spinner disc tip equals approximately:

(a) $\omega \cdot r$ (b) ω/r

(c) r/ω (d) $\omega^2 r$

Q133. Recommended fertilizer band position relative to seed:

(a) In contact (b) 5 cm to the side and 5 cm below

(c) 15 cm above (d) On the surface only

Q134. Direct seed–fertilizer contact is avoided because:

(a) Blocking tubes
(b) Salt injury to germinating seed
(c) Cost
(d) Corrosion of opener

Q135. According to Kepner, the chief benefit of row-crop cultivation is:

(a) Aeration
(b) Moisture mulch
(c) Weed control
(d) Root pruning

Q136. The included angle of cultivator sweeps is generally:

(a) 20–30°
(b) 60–90°
(c) 100–120°
(d) 150°

Q137. Sweeps for dryland moisture conservation are designed with:

(a) High lift and pitch
(b) Low pitch/lift — cut weeds without inversion
(c) Deep suction
(d) Narrow wings

Q138. Near young row crops, cultivation depth should be:

(a) 3–5 cm
(b) 10–12 cm
(c) 15–20 cm
(d) 25 cm

Q139. Spring-trip cultivator standards are set to trip at about:

(a) Working draft
(b) 1.5–2× working draft, with auto-reset
(c) 5× draft
(d) Any load

Q140. Shields on row-crop cultivators:

(a) Reduce draft
(b) Protect young plants from soil throw
(c) Meter seed
(d) Firm soil

Q141. The parallel-linkage gang mounting on cultivators keeps:

(a) Speed constant
(b) Sweep pitch/depth constant over uneven ground
(c) Row spacing fixed
(d) Draft zero

Q142. The rotary hoe is used for:

(a) Deep tillage
(b) Blind (pre/early post-emergence) shallow cultivation at high speed
(c) Puddling
(d) Ridging

Q143. Number of drive-wheel revolutions to simulate area A in drill calibration:

(a) $A/(\pi DWn)$
(b) $\pi DWn/A$
(c) $A \cdot \pi DWn$
(d) $\pi D/AWn$

Q144. Acceptable coefficient of variation between seed-tube discharges of a drill is about:

(a) ≤5–7%
(b) 15–20%
(c) 25–30%
(d) 50%

Q145. A drill with 9 rows at 20 cm spacing, wheel dia 1.2 m: revolutions for 1/25 ha (400 m²) ≈

(a) 25
(b) 59
(c) 100
(d) 236

Answers

Q111 — (c) 95–100%
Q112 — (b) Decreases
Q113 — (b) 50 fpm
Q114 — (b) Use more cells per plate or a larger plate
Q115 — (b) Cutoff device

Q116 — (b) Knock-out/ejector devices
Q117 — (b) Most uniform spacing and minimum seed damage
Q118 — (b) Exposed flute length
Q119 — (b) Steadier, more uniform seed flow
Q120 — (a) $\pi D_w(1-\sigma)/(n_c \cdot i)$
Q121 — (b) Wider than theoretical
Q122 — (b) Greater than the angle of repose of the material
Q123 — (b) 7–10°
Q124 — (b) No-till/zero-till openers
Q125 — (b) Firm soil around seed for better seed–soil contact
Q126 — (b) Hygroscopic, corrosive, variable in flowability
Q127 — (b) Star-wheel feed
Q128 — (b) Gate opening and wheel size/speed
Q129 — (a) Screw speed and pitch volume
Q130 — (b) Triangular/oval, requiring 25–35% overlap
Q131 — (b) Fall short — causing pattern segregation
Q132 — (a) $\omega \cdot r$
Q133 — (b) 5 cm to the side and 5 cm below
Q134 — (b) Salt injury to germinating seed
Q135 — (c) Weed control
Q136 — (b) 60–90°
Q137 — (b) Low pitch/lift — cut weeds without inversion
Q138 — (a) 3–5 cm
Q139 — (b) 1.5–2× working draft, with auto-reset
Q140 — (b) Protect young plants from soil throw
Q141 — (b) Sweep pitch/depth constant over uneven ground
Q142 — (b) Blind (pre/early post-emergence) shallow cultivation at high speed
Q143 — (a) $A/(\pi DWn)$
Q144 — (a) ≤5–7%
Q145 — (b) 59 — *$N = A/(\pi \cdot D \cdot W \cdot n) = 400/(\pi \times 1.2 \times 0.20 \times 9) = 400/6.79 \approx 59$ revolutions.*

Harvester and thresher design — Topic 8 (Q146–Q180)

Q146. In a conventional mower cutter bar, knife-section pitch, guard spacing and stroke are all equal to:

(a) 50.8 mm
(b) 76.2 mm
(c) 101.6 mm
(d) 152.4 mm

Q147. Typical mower crank speed is 800–1000 rpm; the corresponding cutting strokes per minute are:

(a) 400–500
(b) 800–1000
(c) 1600–2000
(d) 3200–4000

Q148. Knife velocity in a crank-driven cutter bar is maximum at:

(a) Start of stroke
(b) End of stroke
(c) Mid-stroke
(d) Uniform throughout

Q149. For 76.2-mm stroke at 900 rpm crank speed, peak knife velocity is about:

(a) 1.2 m/s
(b) 2.3 m/s
(c) 3.6 m/s
(d) 7.2 m/s

Q150. A knife is in proper REGISTER when knife sections centre on the guards:

(a) At mid-stroke
(b) At each end of the stroke
(c) Only while cutting
(d) At maximum velocity

Q151. 'Lead' given to the outer end of a cutter bar compensates for:

(a) Wheel slip
(b) Rearward flexing of the bar under cutting load
(c) Crop moisture
(d) Pitman wear

Q152. Serrations on ledger plates mainly serve to:

(a) Reduce weight
(b) Increase friction so stalks do not slip off the edges
(c) Sharpen the knife
(d) Reduce noise

Q153. Rotary (disc) mowers cut by:

(a) Shear between two edges
(b) Impact/free cutting using stalk inertia
(c) Tearing
(d) Sawing

Q154. Impact cutting is possible only above a:

(a) Minimum stalk diameter
(b) Minimum knife velocity
(c) Maximum moisture
(d) Minimum stubble height

Q155. The wire-loop cylinder is characteristic of:

(a) Maize sheller
(b) 'Hold-on' paddy threshers
(c) Groundnut decorticator
(d) Sunflower thresher

Q156. A rasp-bar cylinder threshes mainly by:

(a) Pure shear
(b) Impact plus rubbing of crop between bars and concave
(c) Centrifuging
(d) Suction

Q157. Cylinder peripheral speed is given by:

(a) $\pi DN/60$
(b) $60/\pi DN$
(c) $\pi D/60N$
(d) $DN/60$

Q158. For wheat threshing at 28 m/s with a 55-cm cylinder, required rpm ≈

(a) 490
(b) 720
(c) 972
(d) 1250

Q159. Raising cylinder speed at fixed clearance generally:

(a) Reduces threshing and damage
(b) Increases threshing completeness AND seed damage
(c) Increases losses only
(d) Has no effect

Q160. Cylinder–concave clearance is normally:

(a) Uniform
(b) Larger at feed end, tapering smaller at rear
(c) Smaller at feed end
(d) Zero at entry

Q161. Visible seed damage rises approximately with:

(a) $\sqrt{v}$
(b) v
(c) v^2
(d) v^3 of cylinder peripheral speed

Q162. For threshing easily damaged podded seeds (e.g., beans), the gentlest unit is:

(a) Spike-tooth cylinder
(b) Rubber rolls at 1.3–1.5 m/s
(c) Hammer beater
(d) High-speed rasp bar

Q163. Threshing efficiency with 2% unthreshed grain is:

(a) 2%
(b) 96%
(c) 98%
(d) 100%

Q164. The axial-flow thresher achieves high efficiency with low damage because:

(a) Very high speed
(b) Crop receives many gentler impacts while spiralling around the cylinder
(c) No concave
(d) Larger seeds

Q165. Indian standard governing thresher feeding-chute safety is:

(a) IS 4468
(b) IS 9129
(c) IS 6284
(d) IS 2062

Q166. In Indian wheat threshers, fine bhusa is produced chiefly by:

(a) Rubber rolls
(b) Beater/'drummy' and spike-tooth types
(c) Wire loop
(d) Stripper header

Q167. In a combine, most of the threshed grain first separates at the:

(a) Straw walkers
(b) Concave and grate
(c) Chaffer sieve
(d) Grain tank

Q168. Straw walkers recover:

(a) Unthreshed heads only
(b) Free grain remaining in the straw mat
(c) Chaff
(d) Weed seed

Q169. The tailings elevator delivers tailings to the:

(a) Grain tank
(b) Cylinder for re-threshing
(c) Straw spreader
(d) Fan

Q170. The upper adjustable-lip sieve of the cleaning shoe is the:

(a) Concave
(b) Chaffer
(c) Scalper
(d) Grader

Q171. Excessive fan blast in the cleaning shoe causes:

(a) Chaffy grain
(b) Free seed blown over the shoe (shoe loss)
(c) Unthreshed heads
(d) Header loss

Q172. Reel index is the ratio of:

(a) Reel diameter to cutting width
(b) Reel peripheral speed to forward speed
(c) Reel rpm to cylinder rpm
(d) Crop height to reel height

Q173. Recommended reel index for combining standing cereals is about:

(a) 0.5–0.8
(b) 1.0
(c) 1.25–1.5
(d) 2.5–3.0

Q174. Too high a reel speed causes mainly:

(a) Unthreshed loss
(b) Grain shattering at the header
(c) Walker loss
(d) Chaffy sample

Q175. A pickup reel (spring-tine, cam-controlled) is used for:

(a) Tall standing crop
(b) Lodged/down crop
(c) Root crops
(d) Green fodder

Q176. At 5 km/h with reel index 1.3 and 1.0-m reel diameter, reel speed ≈

(a) 15 rpm (b) 35 rpm
(c) 70 rpm (d) 140 rpm

Q177. 'MOG' in combine capacity ratings means:

(a) Moisture of grain (b) Material other than grain
(c) Mass of grain (d) Margin of grain

Q178. Unthreshed seed discharged with straw from a combine is classified as:

(a) Header loss (b) Cylinder loss
(c) Shoe loss (d) Rack loss

Q179. In shattering-prone crops, the largest share of total combine loss is usually the:

(a) Header (gathering) loss (b) Cylinder loss
(c) Shoe loss (d) Leakage

Q180. Nearly all seed DAMAGE in a combine occurs at the:

(a) Reel (b) Threshing (cylinder–concave) unit
(c) Grain tank (d) Straw spreader

Answers

Q146 — (b) 76.2 mm

Q147 — (c) 1600–2000 — *Each crank revolution gives two cutting strokes (left + right), so 800–1000 rpm → 1600–2000 strokes/min.*

Q148 — (c) Mid-stroke

Q149 — (c) 3.6 m/s — *$\omega = 2\pi \times 900/60 = 94.2$ rad/s; $r = 38.1$ mm → $v_{max} = \omega r \approx 3.6$ m/s. Velocity is zero at stroke ends — hence cutting is timed near mid-stroke.*

Q150 — (b) At each end of the stroke

Q151 — (b) Rearward flexing of the bar under cutting load

Q152 — (b) Increase friction so stalks do not slip off the edges

Q153 — (b) Impact/free cutting using stalk inertia

Q154 — (b) Minimum knife velocity

Q155 — (b) 'Hold-on' paddy threshers

Q156 — (b) Impact plus rubbing of crop between bars and concave

Q157 — (a) $\pi DN/60$

Q158 — (c) 972 — *$N = 60v/\pi D = 60 \times 28/(3.1416 \times 0.55) = 972$ rpm.*

Q159 — (b) Increases threshing completeness AND seed damage

Q160 — (b) Larger at feed end, tapering smaller at rear

Q161 — (c) v^2 — *Kinetic energy of impact $\propto v^2$ — Kepner Fig. 17.7 shows damage climbing steeply while unthreshed loss falls as speed rises; the design compromise defines the working speed band.*

Q162 — (b) Rubber rolls at 1.3–1.5 m/s

Q163 — (c) 98% — *Threshing efficiency = 100 - % unthreshed = 98%.*

Q164 — (b) Crop receives many gentler impacts while spiralling around the cylinder

Q165 — (b) IS 9129

Q166 — (b) Beater/'drummy' and spike-tooth types

Q167 — (b) Concave and grate

Q168 — (b) Free grain remaining in the straw mat

Q169 — (b) Cylinder for re-threshing

Q170 — (b) Chaffer

Q171 — (b) Free seed blown over the shoe (shoe loss)

Q172 — (b) Reel peripheral speed to forward speed

Q173 — (c) 1.25–1.5

Q174 — (b) Grain shattering at the header

Q175 — (b) Lodged/down crop

Q176 — (b) 35 rpm — *Reel peripheral speed = 1.3 × (5/3.6) = 1.81 m/s; N = 60v/πD = 60×1.81/3.1416 ≈ 34.5 rpm.*

Q177 — (b) Material other than grain

Q178 — (b) Cylinder loss

Q179 — (a) Header (gathering) loss — *Header/gathering (shatter) loss commonly dominates in crops like sesame, mustard, soybean; it is controlled by reel setting, forward speed and timely harvest.*

Q180 — (b) Threshing (cylinder–concave) unit

Controls, mechatronics, ML, IoT, robotics and drones — Topics 9, 10, 12 (Q181–Q215)

Q181. Hydraulic systems work on:

(a) Bernoulli's theorem
(b) Pascal's law
(c) Archimedes' principle
(d) Boyle's law

Q182. Oil at 200 bar acts on a 50-cm² piston. The force developed is:

(a) 10 kN
(b) 50 kN
(c) 100 kN
(d) 200 kN

Q183. Hydraulic power in kW with p in bar and Q in L/min is:

(a) pQ/60
(b) pQ/367
(c) pQ/600
(d) pQ/1000

Q184. The hydraulic pump almost universally used on agricultural tractors is:

(a) Centrifugal
(b) External gear
(c) Diaphragm
(d) Screw

Q185. In an OPEN-centre hydraulic system, in neutral the pump flow:

(a) Is blocked
(b) Returns freely to tank at low pressure
(c) Goes to the cylinder
(d) Stops

Q186. A closed-centre load-sensing system uses a:

(a) Fixed-displacement pump running at relief pressure
(b) Variable-displacement pump supplying flow on demand
(c) Hand pump
(d) Air compressor

Q187. The relief valve in a hydraulic circuit:

(a) Directs flow
(b) Limits maximum system pressure
(c) Filters oil
(d) Cools oil

Q188. A single-acting hydraulic cylinder is lowered by:

(a) Oil pressure
(b) Gravity/implement weight
(c) Spring only
(d) Air

Q189. In a vacuum precision planter, seeds are held on the metering plate by:

(a) Positive air pressure
(b) Negative pressure (suction)
(c) Glue
(d) Magnetism

Q190. An air seeder distributes seed from a central hopper to openers by:

(a) Gravity tubes
(b) Augers
(c) An air stream through manifolds
(d) Chain conveyors

Q191. Compared with hydraulics, pneumatic systems have:

(a) Higher force density
(b) Lower force density but cleaner operation
(c) Same force
(d) No actuators

Q192. Mechatronics is the integration of:

(a) Only mechanics and hydraulics
(b) Mechanics, electronics and computing/control
(c) Chemistry and biology
(d) Civil and electrical works

Q193. The international standard for the tractor–implement electronic interface (ISOBUS) is:

(a) ISO 730
(b) ISO 11783
(c) ISO 9001
(d) ISO 500

Q194. ISOBUS communication is based on:

(a) Bluetooth
(b) CAN bus
(c) Wi-Fi
(d) RS-232 only

Q195. Draft on modern electro-hydraulic hitch tractors is sensed by:

(a) Spring balance
(b) Load-sensing pins in the lower links
(c) Operator feel
(d) Fuel gauge

Q196. Supervised machine learning requires:

(a) Unlabelled data
(b) Labelled training data
(c) No data
(d) Only images

Q197. The neural-network type dominating image-based weed detection is:

(a) RNN
(b) CNN (convolutional)
(c) Perceptron only
(d) SOM

Q198. 'See & Spray' technology saves chemicals by:

(a) Diluting spray
(b) Spraying only where ML vision detects weeds
(c) Spraying at night
(d) Using more nozzles

Q199. The IoT protocol designed for kilometres of range at very low power is:

(a) Bluetooth
(b) LoRaWAN
(c) USB
(d) NFC

Q200. NDVI is computed as:

(a) (R-NIR)/(R+NIR)
(b) (NIR-R)/(NIR+R)
(c) NIR×R
(d) NIR/G

Q201. RTK-GNSS auto-steer gives pass-to-pass accuracy of about:

(a) 2–3 m
(b) 30–50 cm
(c) 2–3 cm
(d) 2–3 mm

Q202. The robot component that actually grips/cuts the fruit is the:

(a) Manipulator base
(b) End-effector
(c) Encoder
(d) Gantry

Q203. Edge computing in IoT means processing data:

(a) Only in cloud
(b) On or near the sensing device
(c) On paper
(d) After harvest

Q204. Under Drone Rules 2021, agricultural spray drones (typically 2–25 kg) fall in the class:

(a) Nano
(b) Micro
(c) Small
(d) Medium

Q205. A NANO drone weighs:

(a) <250 g (b) <2 kg

(c) <25 kg (d) <150 kg

Q206. India's online platform for drone registration, UIN and airspace maps is:

(a) UMANG (b) Digital Sky (DGCA)

(c) DigiLocker (d) AGMARKNET

Q207. In a GREEN zone, drones may operate without permission up to:

(a) 60 m (b) 120 m (400 ft)

(c) 500 m (d) Unlimited

Q208. A Remote Pilot Certificate is NOT required for:

(a) Small drones (b) Medium drones

(c) Nano and non-commercial micro drones (d) Any drone

Q209. Typical drone ULV spray application rate is:

(a) 2–4 L/ha (b) 20–40 L/ha

(c) 200–400 L/ha (d) 500–600 L/ha

Q210. Recommended spraying height of an agri drone above the crop canopy is about:

(a) 0.2–0.5 m (b) 2–3 m

(c) 8–10 m (d) 15–20 m

Q211. Drone spraying should be avoided when wind speed exceeds about:

(a) 2 km/h (b) 10 km/h

(c) 30 km/h (d) 50 km/h

Q212. Agricultural drones are commonly powered by:

(a) Lead-acid batteries (b) LiPo (lithium-polymer) batteries

(c) NiCd (d) Dry cells

Q213. Under SMAM, ICAR institutes/KVKs/SAUs get drone assistance of:

(a) 40% (b) 50%

(c) 75% (d) 100% up to ₹10 lakh

Q214. Namo Drone Didi provides drones to women SHGs at a subsidy of:

(a) 40% up to ₹4 lakh (b) 50% up to ₹5 lakh

(c) 80% up to ₹8 lakh (d) 100%

Q215. For large-area aerial survey/imaging with long endurance, the preferred platform is:

(a) Quadcopter (b) Fixed-wing drone

(c) Tethered balloon (d) Hexacopter

Answers

Q181 — (b) Pascal's law

Q182 — (c) 100 kN — *$F = pA = 200\times10^5$ Pa × 50×10^{-4} m² = 100,000 N = 100 kN — hydraulic force multiplication.*

Q183 — (c) pQ/600 — *P(kW) = p(bar)×Q(L/min)/600; e.g., 180 bar × 40 L/min → 12 kW.*

Q184 — (b) External gear

Q185 — (b) Returns freely to tank at low pressure

Q186 — (b) Variable-displacement pump supplying flow on demand

Q187 — (b) Limits maximum system pressure

Q188 — (b) Gravity/implement weight

Q189 — (b) Negative pressure (suction) — *Vacuum plate = singulation by suction (precision planting); air seeder = pneumatic bulk distribution — two different uses of air.*

Q190 — (c) An air stream through manifolds — *Vacuum plate = singulation by suction; air seeder = pneumatic bulk distribution — two different uses of air.*
Q191 — (b) Lower force density but cleaner operation
Q192 — (b) Mechanics, electronics and computing/control
Q193 — (b) ISO 11783
Q194 — (b) CAN bus
Q195 — (b) Load-sensing pins in the lower links
Q196 — (b) Labelled training data
Q197 — (b) CNN (convolutional)
Q198 — (b) Spraying only where ML vision detects weeds
Q199 — (b) LoRaWAN
Q200 — (b) (NIR-R)/(NIR+R) — *Healthy vegetation reflects NIR strongly and absorbs red → NDVI near +0.6–0.9; stressed crop lower (basis of drone crop monitoring, Unit 10).*
Q201 — (c) 2–3 cm
Q202 — (b) End-effector
Q203 — (b) On or near the sensing device
Q204 — (c) Small — *Drone Rules 2021 — nano <250 g, micro ≤ 2 kg, small ≤ 25 kg, medium ≤ 150 kg, large >150 kg.*
Q205 — (a) <250 g
Q206 — (b) Digital Sky (DGCA)
Q207 — (b) 120 m (400 ft) — *Green zone ceiling 120 m.*
Q208 — (c) Nano and non-commercial micro drones — *RPC exemption only for nano and non-commercial micro drones.*
Q209 — (b) 20–40 L/ha
Q210 — (b) 2–3 m
Q211 — (b) 10 km/h
Q212 — (b) LiPo (lithium-polymer) batteries
Q213 — (d) 100% up to ₹10 lakh — *SMAM slabs — 100%/₹10 lakh (ICAR/KVK/SAU), 75% FPO demos, CHC 40%/₹4 lakh (50%/₹5 lakh agri graduates), individuals 40–50%.*
Q214 — (c) 80% up to ₹8 lakh — *Namo Drone Didi — 80%/₹8 lakh, 15,000 women SHGs, ₹1,261 crore (2023-24 to 2025-26).*
Q215 — (b) Fixed-wing drone